DRAMANE DIARRA

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PROFILE

PhD-level researcher in machine learning and large-scale optimization, specializing in reinforcement learning, simulation-based optimization, and data-driven decision-making. Experienced in designing scalable algorithms for complex, high-dimensional and multi-agent systems under uncertainty. My work focuses on non-convex optimization and sequential decision-making in distributed environments, with a particular interest in hybrid approaches combining learning-based methods with classical optimization to build robust and interpretable decision-support systems.

RESEARCH INTERESTS

- Deep Reinforcement Learning
- Large-scale optimization
- Machine Learning for Energy-Efficient IoT Networks
- Large language Models (LLMs) and LLM-assisted optimization
- Generative AI
- Learning-Based Resource Allocation in Wireless Systems
- Distributed and Edge Intelligence for Communication Networks
- RF/microwave, antennas

EDUCATION

Pan African University for basic science, technology, and innovation (PAUSTI) **Nairobi, Kenya**
PhD. in Electrical Engineering (Telecommunications option) April 2023 – April 2026
Topic: Deep Reinforcement Learning-based power control, and power allocation for maximizing Energy Efficiency in User-centric Cell Free massive MIMO networks.

University 20 August 1955, Skikda **Skikda, Algeria**
Master’s Degree in Electrical Engineering (Telecommunications systems) November 2019 -July 2021

University 20 August 1955, Skikda **Skikda, Algeria**
Bachelor’s Degree in Telecommunications November 2016 -July 2019

RESEARCH EXPERIENCE

Pan African University for basic science, technology, and innovation (PAUSTI) **Nairobi, Kenya**
PhD Researcher 2023 - 2026

- Designed and developed deep reinforcement learning algorithms for energy-efficient resource allocation in user-centric cell-free massive MIMO networks.
- Developed Backhaul and fronthaul-aware resource strategies in user-centric cell-free massive MIMO networks for heterogenous users demands.
- Authored and co-authored peer-reviewed journal articles on AI-driven wireless network optimization and energy-efficient communication systems.
- Collaborated with supervisors and international researchers on the design, validation, and dissemination of learning-based optimization techniques for next-generation (5G/6G) networks.

Jomo Kenyatta University of Agriculture and Technology (JKUAT) **Nairobi, Kenya**
Research / Part-time Lecturer 2023 - 2026

- Conducted applied research on signal processing and optimization techniques for wireless communication systems alongside doctoral studies.
- Supervised undergraduate research projects and capstone designs related to wireless networks, signal processing, and embedded systems.
- Mentored students on simulation-based research methodologies.
- Contributed to curriculum development by integrating recent research advances in machine learning and wireless communications into teaching materials.

University 20 August 1955, Skikda **Skikda, Algeria**
Graduate Researcher (Master) 2019 - 2021

- Designed and simulated a 2×2 dual-band patch MIMO antenna operating at 28/38 GHz for 5G mobile communications.
- Evaluated antenna performance in terms of isolation, gain, bandwidth, and radiation patterns using electromagnetic simulation tools.
- Analyzed the impact of antenna design choices on system-level performance and energy efficiency.
- Analyzed the compatibilities of electromagnetic

RESEARCH PUBLICATIONS

Published:

1. **Dramane Diarra**, Heywood Ouma Absaloms, & Philip Kibet Langat.. (2026). Optimizing Uplink Power Control for Energy Efficiency in mmWave User-Centric Cell-Free Massive MIMO with Deep Reinforcement Learning. Journal of Engineering and Applied Science, vol. 73, Article no. 11, published 09 January 2026. Publisher: Springer Nature (SCI indexed). DOI: 10.1186/s44147-025-00864-w.

2. **Dramane Diarra**, Heywood Ouma Absaloms, & Philip Kibet Langat.. (2025). Deep Reinforcement Learning-based Power Allocation for Maximizing Energy Efficiency in mmWave Cell-free Massive MIMO. International Journal of Intelligent Engineering and Systems, 18(10), 450-465. Publisher: INASS (SCI indexed). DOI: 10.22266/ijies2025.1130.29

Under review:

3. **Dramane Diarra**, Heywood Ouma Absaloms, & Philip Kibet Langat. Energy-Efficient Downlink Power Allocation in User-Centric Cell-Free Massive MIMO Networks Using Hybrid Deep Reinforcement Learning Algorithm.
IEEE Transactions on Communications (under review). Publisher: IEEE (SCI indexed)

4. **Dramane Diarra**, Heywood Ouma Absaloms, & Philip Kibet Langat. Multi-Agent Deep Reinforcement Learning-Based Power Control for Energy-Efficient User-Centric Cell-Free Massive MIMO. Physical Communication (Under review). Publisher: Elsevier (SCI indexed)

5. **Dramane Diarra**, Heywood Ouma Absaloms, & Philip Kibet Langat. Deep Reinforcement Learning-based Power Control Optimization for Maximizing Energy Efficiency in mmWave User-Centric Cell-Free Massive MIMO. IET Communications (Under review). Publisher: Wiley & Son (SCI indexed)

WORK EXPERIENCE

Jomo Kenyatta University of Agriculture and Technology (JKUAT) — **Nairobi, Kenya**
Part-time Lecturer — 2023 - 2025
Department of Telecommunication and Information Engineering
Deliver undergraduate instruction in signal processing and digital electronics through lectures, laboratory sessions, and assessments.

IMPERIS Sarl — **Bamako, Mali**
Engineer — December 2022 – April 2023
- Optical network engineering and microwave link installation
- Installation and configuration of IP cameras

École Relationnelle d'Informatique et Électronique — **Bamako, Mali**
Part-time Lecturer — December 2022 – March 2023
- Taught analog and digital transmission systems to undergraduate students.
- Delivered comprehensive instruction in network planning methodologies and design principles.

Orange Mali — **Bamako, Mali**
Technical Intern — September 2022 - November 2022
Enterprise and Residential Network Engineering Division
- Supported FTTX deployment and configuration for enterprise and residential networks.
- Participated in wireless fixed hub network implementation projects.
- Assisted back-office operations and network engineering documentation processes.

École Universitaire de Technologies et de Gestion (EUTG) — **Bamako, Mali**
Part-time Lecturer — January 2022 – August 2022
- Taught algorithm and programming with emphasis on Java fundamentals.
- Designed coding exercises and assessed student programming competencies through practical evaluations.

Institut Supérieur des Affaires et de Technologies (ISPATEC) — **Bamako, Mali**
Part-time Lecturer — February 2021 – March 2023
- Delivered instruction in fundamental electronics, digital electronics, Windows Server, Linux systems, and Optical fiber to undergraduate students,
- Conducted practical demonstrations and supervised hands-on laboratory work.

INTEC-Sup — **Bamako, Mali**
Part-time Lecturer — April 2021 – July 2022
- Taught digital electronics, network management, and Linux system administration.
- Integrated theoretical knowledge with practical system configurations and real-world applications.

TECHNICAL SKILLS

- Technical Report Writing
- Python, R, C++, Java, C#, C, JavaScript, .NET
- Machine Learning, Deep Learning
- Reinforcement Learning

- Large Language Models (LLMs), Generative AI
- Natural Language Processing (NLP)
- Data Analysis and Visualization
- Power BI, SQL, Excel
- Optimization
- TCP/IP Fundamentals
- Routing and Switching
- DNS, HTTP, DHCP
- Network Simulation
- MATLAB, Ansys HFSS, CST, COMSOL
- Linux/Unix environments, Shell scripting

CERTICFIATION

English Language Certification (B2 - Independent User) 2020
Intensive Languages Teaching Centre at University of Skikda, Algeria

Learning AI Through Visualization April to July 16, 2025
Colombia university

English Language Level – Intermediate April 1, 2023 – August 20, 2023
Pan African University Institute for Basic Sciences, Technology, and Innovation

LANGUAGE

- English: Full Professional proficiency
- French: Native
- Bambara: native
- Kiswahili: Beginner
- Arabic: Beginner

REFRENCES

Prof. Heywood Ouma Absaloms, PhD Supervisor, University of Nairobi
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Prof. Philip Kibet Langat, PhD Supervisor, Jomo Kenyatta University of Agriculture and Technology
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